YUEZHAO LI

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Personality

Nationality Chinese
Date of birth 18 April, 1996
Languages Chinese (Native), English (C1), German (B1), Dutch (A1)

Research interest

- ◇ Noncommutative geometry via spectral triples and unbounded Kasparov modules.
- ◇ Spectral localisers and computational (bivariant) K-theory of C*-algebras.
- ◇ Topological phases of matter via groupoids and coarse geometry.

Appointments

Max-Planck-Institut für Mathematik Mar 2026 — Aug 2026
Visiting postdoctoral researcher *Germany*
- ◇ Host: Prof. Dr. Matthias Lesch.

Education

Universiteit Leiden Oct 2021 — Feb 2026
PhD in mathematics *The Netherlands*
- ◇ Supervisor: Dr. Bram Mesland.
- ◇ Thesis: *Spectral localisers and aperiodic topological phases in noncommutative geometry.*

Georg-August-Universität Göttingen Oct 2018 — Aug 2021
Master in mathematics, with minor in physics *Germany*
- ◇ Thesis advisor: Prof. Dr. Ralf Meyer.
- ◇ Thesis: *Invariants for topological insulators coming from decompositions of coarse spaces.*

Peking University Sep 2013 — Jul 2018
Bachelor in physics, with minor in mathematics *China*
- ◇ Thesis advisor: Prof. Dr. Limei Xu.
- ◇ Thesis: *Phase transition and critical phenomena.*

Publications

Peer-reviewed journal articles

[1] Y. Li and B. Mesland. *The odd spectral localiser via asymptotic morphisms and quasi-projections.* Ann. K-Theory **11**, no. 2 (2026), 213–260. doi: 10.2140/akt.2026.11.213.

[2] Y. Li. *Robustness of Topological Phases on Aperiodic Lattices.* Math. Phys. Anal. Geom. **29**, no. 2 (2026), Paper No. 16. doi: 10.1007/s11040-026-09547-1.

Preprints

[3] Y. Li and G. C. Thiang. *Consistent symmetry breaking and topological phases* (2026). arXiv: 2607.28181.

Talks

Research talks

- ◇ October 2026: *Symmetry breaking colimit of topological phases*.
  Noncommutative geometry in NYC. Online seminar.
- ◇ September 2026: *Consistent symmetry breaking and topological phases*.
  K-theory of C*-algebras in solid state physics: 40 years later, in honor of Jean Bellissard's 80th birthday. Potsdam, Germany.
- ◇ August 2026: *Symmetry breaking colimit of topological phases and equivariant coarse indices*.
  NoSeaG 2026. Prague, Czech Republic.
- ◇ July 2026: *The spectral localiser via* E*-theory*.
  Frontiers in Noncommutative Geometry and its Applications. Hangzhou, China.
- ◇ May 2026: *Noncommutative geometry of topological phases of matter*.
  Higher Differential Geometry Seminar. Max-Planck-Institut für Mathematik, Germany.
- ◇ April 2026: *The spectral localiser: what and why*.
  Oberseminar Global Analysis and Operator Algebras. Universität Bonn, Germany.
- ◇ December 2025: *The spectral localiser via* E*-theory*.
  NCG Colloquim II. Nijmegen, Netherlands.
- ◇ November 2025: *The spectral localiser via* E*-theory*.
  Oberseminar Global Analysis and Operator Algebras. Universität Bonn, Germany.
- ◇ August 2025: *On topological phases of aperiodic matter*.
  NCG Day. Leiden, Netherlands.
- ◇ July 2025: *The spectral localiser via* E*-theory*.
  Oberseminar Analysis. Leibniz Universität Hannover, Germany.
- ◇ July 2025: *The comparison maps between groupoid and Roe* C*-*algebras from aperiodic topological insulators*.
  Oberseminar Noncommutative Geometry. Georg-August-Universität Göttingen, Germany.
- ◇ June 2025: *The spectral localiser via* E*-theory*.
  C*-algebras, Coarse Geometry and Physics. Greifswald, Germany.
- ◇ April 2025: *On robustness of topological phases* (short talk).
  Applications of Noncommutative Geometry to Gauge Theories, Field Theories, and Quantum Spacetime. Marseille, France.
- ◇ March 2025: *On robustness of topological phases of matter*.
  Seminar in Analysis and Applications. Delft University of Technology, Netherlands.
- ◇ January 2025: *Understanding strong and weak topological phases*.
  Forschungsseminar Algebra/Geometrie/Topologie. Universität Greifswald, Germany.
- ◇ September 2024: *Index theory of spectral truncations and the localisation algebra*.
  East China Normal University, Shanghai, China.
- ◇ October 2023: *Models of aperiodic topological insulators* (short talk).
  Autumn School on Large-scale Geometry. Göttingen, Germany.

Colloquium talks

- ◇ July 2026: K*-theory in solid-state physics: modelling and computation*.
  MPI-Oberseminar, Bonn, Germany.
- ◇ September 2024: *Noncommutative geometry of integer quantum Hall effect*.
  Sichuan University, Chengdu, China.

- ◇ May 2024: *A hitchhiker's guide to topological insulators in noncommutative geometry.* Leiden–Birmingham meeting, Leiden, Netherlands.
- ◇ November 2023: *Noncommutative geometry in index theory and physics.* Leiden–Luxembourg PhD Away Day. Leiden, Netherlands.

Conferences, workshops and seminars

Participation

- ◇ September 2026: K-theory of C*-algebras in solid state physics: 40 years later, in honor of Jean Bellissard's 80th birthday. Potsdam, Germany.
- ◇ September 2026: Interactions between operator algebras, K-theory and homotopy theory. Münster, Germany.
- ◇ August 2026: NoSeaG 2026. Prague, Czech Republic.
- ◇ July 2026: Summer School and International Workshop on Noncommutative Geometry. Dalian, China.
- ◇ July 2026: Frontiers in Noncommutative Geometry and its Applications. Hangzhou, China.
- ◇ June 2026: Interfaces of Spectral Theory, Operator Algebras and Noncommutative Geometry, on the occasion of Matthias Lesch's 65 birthday. Bonn, Germany.
- ◇ December 2025: NCG Colloquim II. Nijmegen, Netherlands.
- ◇ August 2025: NCG Day. Leiden, Netherlands.
- ◇ June 2025: C*-algebras, Coarse Geometry and Physics. Greifswald, Germany.
- ◇ April 2025: Applications of Noncommutative Geometry to Gauge Theories, Field Theories, and Quantum Space-time. Marseille, France.
- ◇ October 2024: Noncommutativity behind the Dunes. Delft, Netherlands.
- ◇ August 2024: OdenSeaG 2024. Odense, Denmark.
- ◇ May 2024: Leiden–Birmingham meeting. Leiden, Netherlands.
- ◇ April 2024: Group Operator Algebras and Noncommutative Geometry. Marseille, France.
- ◇ November 2023: Leiden–Luxembourg PhD Away Day. Leiden, Netherlands.
- ◇ October 2023: Autumn School on Large-scale Geometry. Göttingen, Germany.
- ◇ August 2023: YMC*A. Leuven, Belgium.
- ◇ May 2023: Noncommutative Geometry and Operator Algebras (NSeaG 2023). Bonn, Germany.
- ◇ March 2023: Noncommutative Geometry and Representation Theory. Athens, Greece.
- ◇ March 2022: Noncommutativity in the North. Gothenburg, Sweden.
- ◇ February 2022: Winter School on Foliations, Pseudodifferential Operators and Groupoids. Göttingen, Germany.

Organisation

- ◇ January 2023: 5th Conference of Settat on Operator Algebras and Applications. Co-organiser, with Francesca Arici, Marcel de Jeu, Rachid El Harti and Dimitris Gerontogiannis.
- ◇ I organised several Leiden Local NCG Seminars and took notes for them.
  - Spring 2022: KK-theory (Notes).
  - Fall 2022: Groupoid C*-algebras (Notes).
  - Fall 2023: Noncommutative geometry of foliations (Notes).

Research visits

- ◇ September 2026: Universiteit Leiden, Netherlands. Hosted by Dr. Bram Mesland.
- ◇ December 2025: Oberwolfach, Germany. Hosted by Dr. Jialong Deng.
- ◇ November 2025: Max Planck Institute for Mathematics, Germany. Hosted by Dr. Malte Leimbach.
- ◇ July 2025: Leibniz Universität Hannover, Germany. Hosted by Dr. Eske Ewert.

- July 2025: Georg-August-Universität Göttingen, Germany. Hosted by Prof. Dr. Ralf Meyer.
- January 2025: Universität Greifswald, Germany. Hosted by Prof. Dr. Matthias Ludewig.
- October 2024: Peking University, China. Hosted by Dr. Guo Chuan Thiang.
- September 2024: East China Normal University, China. Hosted by Prof. Dr. Hang Wang.
- September 2024: Sichuan University, China. Hosted by Dr. Xingni Jiang and Dr. Yunfeng Shi.

Teaching experience

Universiteit Leiden

- Fall 2024: Functional Analysis, Mastermath course. Teaching assistant.
- Spring 2024: Operator Algebras, Mastermath course. Teaching assistant.
- Fall 2023: Functional Analysis, Mastermath course. Teaching assistant.
- Spring 2023: Differentiable Manifolds II, undergraduate course. Teaching assistant.
- Fall 2022: Functional Analysis, Mastermath course. Teaching assistant.
- Spring 2022: Differentiable Manifolds II, undergraduate course. Teaching assistant.
- Fall 2021: Functional Analysis, Mastermath course. Teaching assistant.